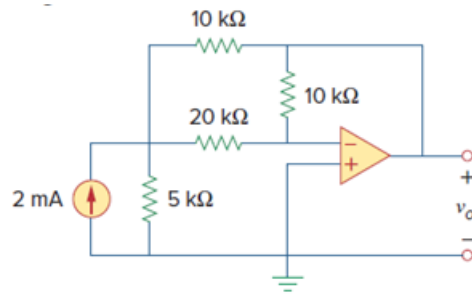


Problem

Determine the output voltage v_o in the circuit of Fig. 5.53.

Figure 5.53:



Step-by-step solution

Step 1 of 4

Refer to the circuit diagram in Figure 5.52 in the textbook.

Consider operational amplifier is an ideal op-amp.

Modify circuit diagram is as shown in Figure 2.

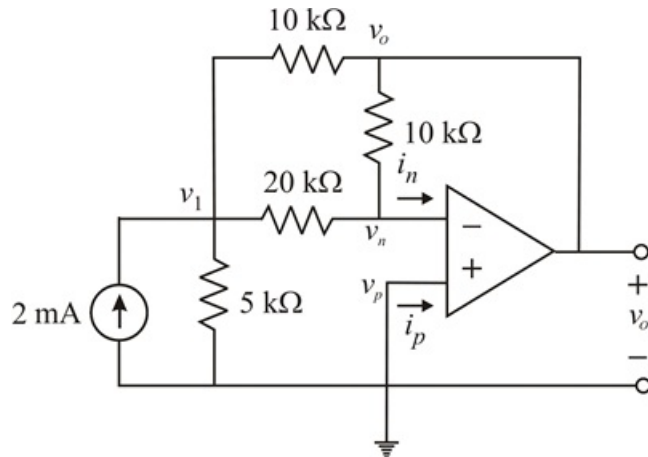


Figure 2

Step 2 of 4

Due to virtual short-circuit between inverting and non-inverting terminals of the op-amp

$$\begin{aligned}v_p &= v_n \\&= 0 \text{ V}\end{aligned}$$

The entering current in the inverting and the non-inverting terminal is zero.

$$\begin{aligned}i_p &= i_n \\&= 0 \text{ A}\end{aligned}$$

Apply Kirchhoff's current law at node v_1 .

$$\begin{aligned}2\text{m} &= \frac{v_1}{5\text{k}} + \frac{v_1 - v_n}{20\text{k}} + \frac{v_1 - v_o}{10\text{k}} \\40 &= 4v_1 + v_1 - v_n + 2v_1 - 2v_o \\7v_1 - v_n - 2v_o &= 40 \\7v_1 - 0 - 2v_o &= 40 \\7v_1 - 2v_o &= 40 \dots\dots (1)\end{aligned}$$

Step 3 of 4

Apply Kirchhoff's current law at node v_n .

$$\begin{aligned}\frac{v_n - v_1}{20\text{k}} + \frac{v_n - v_o}{10\text{k}} + i_n &= 0 \\ \frac{0 - v_1}{20\text{k}} + \frac{0 - v_o}{10\text{k}} + 0 &= 0 \\ \frac{-v_1}{20\text{k}} + \frac{-v_o}{10\text{k}} &= 0 \\ -v_1 - 2v_o &= 0 \\ v_1 &= -2v_o\end{aligned}$$

Step 4 of 4

Recall equation (1).

$$7v_1 - 2v_o = 40$$

Substitute $-2v_o$ for v_1 in the equation.

$$\begin{aligned}7(-2v_o) - 2v_o &= 40 \\ -14v_o - 2v_o &= 40 \\ -16v_o &= 40 \\ v_o &= \frac{-40}{16} \\ &= -2.5\end{aligned}$$

Thus, the output voltage v_o is $\boxed{-2.5\text{V}}$.